Review Report

Murex, Quark: Convexity Adjustments

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1 Executive Summary

Conclusions For the pricing of financial derivatives, commonly expectations of stochastic variables (e.g., stochastic rates) have to be calculated. The stochastic variable (or underlying) has a corresponding underlying measure, often referred to as the natural measure. If the expectation of a stochastic variable is calculated under a different measure than the natural measure, this gives rise to a convexity adjustment. For the pricing of financial derivatives, neglecting convexity adjustments can lead to incorrect prices and sensitivities.

For efficiency purposes, it is desirable to have simple numerical calculations, preferably analytical, in order to calculate the Net Present Value (NPV) and sensitivities for derivatives. To do so, often additional approximations are necessary. These additional approximations can only be taken if the required accuracy is still achieved. The type of additional approximations of interest is the approximation of neglecting a convexity adjustment. A convexity adjustment is a non-linear term in the valuation. By neglecting a convexity adjustment, a risk-factor is removed from the valuation, since the impact of volatility and/or correlation is being ignored. However, the impact of doing so can be negligible under specific conditions. When giving sign-off for pricing functions that can be used to calculate the NPV and sensitivities for derivatives these specific conditions are key. Typically, PMV imposes limitations such that these specific conditions are met. Meaning that within the limitations, neglecting a convexity adjustment is justifiable.

PMV always spends considerable time and effort on this important topic during validations and periodic reviews. As such, many analyses and limitations are in place.

As of 2021, PMV maintains an overview with information about the calculated and neglected convexity adjustments, and the associated validation deliverables, that are in place for currently signed-off pricing functions. Every calculated and neglected convexity adjustment should be properly explained within at least one validation report. Furthermore, if required, an impact analysis should be available and proper limitations should be imposed.

The convexity adjustment overview, *ConvexityAdjustments.xlsx* [12], enables PMV to assess where further improvements are necessary. These improvements should in principle be made during validations. However, they can also be made during model category reviews and system level item reviews, such as this one.

The main goal of this periodic review is to keep the convexity adjustment overview up-to-date. Furthermore, this report also explains the importance of convexity adjustments from a general perspective. The focus of this report is not on individual convexity adjustments, it will not contain information about the impact of neglecting each convexity adjustment. This should be reflected by the validation deliverables, the validation report and associated settings tables. Only a high-level summary of this information should be added to the convexity adjustment overview.

During this year's review, it has been checked if the convexity adjustment overview is still up-to-date. The majority of the time has been spend on improving validation reports with respect to (neglected) convexity adjustments and on setting up a new repository which will improve the maintenance in the (re)creation of comprehensive tests.

The list below summarizes the issues encountered during the review and the actions taken by PMV.

Issue 1 It needs to be checked if the convexity adjustment overview is up-to-date for the pricing systems that are in scope.

Action 1 Firstly, Sophis and Luna have been taken out of the list of considered pricing systems that are in scope for PMV since these pricing system are no longer in use. Secondly, the overview has been checked and it turned out that it was not fully up-to-date. For a Murex neglected convexity adjustment related to average floating rate several fields required to be updated as a result of work done in this review. The comprehensive testing done related to this convexity adjustment is more clearly described in the *IRSVanilla* [13] validation report, and this is now also reflected in the fields of the overview. Furthermore, for Quark some minor typos were fixed. The overview has been updated such that it is up-to-date. The issue has been resolved, but needs to be revisited during next year's periodic review.

Issue 2 One neglected convexity adjustment, *No convexity for floating rates with non-standard rate conventions*, deserves special attention as it has not been handled during any validation or periodic review thus far. This item is part of the convexity adjustment overview but it is not explained within any Murex validation report, nor were any limitations imposed.

Action 2 The neglected convexity adjustment, *No convexity for floating rates with non-standard rate conventions*, was already added to the convexity adjustments overview during the periodic review of 2021. For now these explanations and derivations are included in this specific report. However, the idea is to move these explanations and derivations to the validation report of the pricing function *IRSVanilla* [13]. The rewrite of this validation report is a separate project on PMV's project list, Epic 2190248 and is still ongoing. Currently, another validation project is being worked on wherein part of the rewrite will be done, Epic 8426332. However, the expectation is that after this validation more work will still be required as the rewrite will include adding comprehensive testing for some neglected convexity adjustments that could also result in imposing additional limitations. This issue remains open.

Issue 3 The convexity adjustment overview shows that for some convexity adjustments, the validation deliverables are lacking (e.g., explanation in a validation report is incomplete, impact analysis is missing).

Action 3 In the periodic review of 2023 a new repo was created to better maintain all materials related to comprehensive testing. The idea of this repo is that it will be as self-contained as possible, such that it is easy to recreate comprehensive tests. In this year PMV has worked on the design of the repo and put a structure in place. PMV also moved two comprehensive tests to this repo, one for Quark and one for Murex. Moving all comprehensive tests is still a substantial amount of work. The expectation is that with the structure in place this work can now also be done as part of ongoing related validations. The progress for the comprehensive testing repo is tracked via a separate project on PMV's project list, Epic 8192740. Furthermore in this review, PMV moved the performed analysis of the 2022 periodic review on Convexity Adjustment for Average Indices (Murex) to the *IRSVanilla* [13] validation report, and improved the testing section in this validation report. This issue remains open.

Issue 4 The neglecting of convexity adjustments can be viewed as model weaknesses. As per the model risk reserve policy [1], PMV should consider proposing a model risk reserve in case of a model weakness. Currently, no model risk reserves are in place related to these model weaknesses.

Action 4 Most limitations related to convexity adjustments that have been imposed by PMV, which are a mitigating factor, are currently not being checked by Product Control (PC) as far as PMV is aware. Reason being, that PC is not checking market data for Murex and Quark. To mitigate the effect of these violations, a model risk reserve should be taken. PMV planned to create a generic pricing library which will include a generic Monte-Carlo framework. This is a separate project on PMV's project list, Epic 2459425. The hope is that this library can eventually also help with determining whether a model risk reserve is deemed necessary for a particular model. This project is currently on hold and it is not clear if it will be worked on next year as the primary focus is currently on the new comprehensive testing repo, Epic

8192740. This issue remains open.

This concludes the review for 2024. The review will be continued in 2025.

2 Introduction

For the pricing of financial derivatives, commonly expectations of stochastic variables (e.g., stochastic rates) have to be calculated. The stochastic variable (or underlying) has a corresponding underlying measure, often referred to as the natural measure. If the expectation of a stochastic variable is calculated under a different measure than the natural measure, this gives rise to a convexity adjustment. For the pricing of financial derivatives, neglecting convexity adjustments can lead to incorrect prices and sensitivities.

PMV always spends considerable time and effort on this important topic during validations and periodic reviews. As such, many analyses and limitations are in place.

As of 2021, PMV maintains an overview with information about the calculated and neglected convexity adjustments, and the associated validation deliverables, that are in place for currently signed-off pricing functions. Every calculated and neglected convexity adjustment should be properly explained within at least one validation report. Furthermore, if required, an impact analysis should be available and proper limitations should be imposed.

The convexity adjustment overview, *ConvexityAdjustments.xlsx* [12], enables PMV to assess where further improvements are necessary. These improvements should in principle be made during validations. However, they can also be made during model category reviews and system level item reviews, such as this one.

The main goal of this periodic review is to keep the convexity adjustment overview up-to-date. Furthermore, this report also explains the importance of convexity adjustments from a general perspective. The focus of this report is not on individual convexity adjustments, it will not contain information about the impact of neglecting each convexity adjustment. This should be reflected by the validation deliverables, the validation report and associated settings tables. Only a high-level summary of this information should be added to the convexity adjustment overview.

Convexity adjustments are a broader topic that can be relevant for any pricing system. This depends on the pricing models/functions contained within the pricing system. The convexity adjustments review topic is a reoccurring system level periodic review item. Pricing systems are only included in the convexity adjustment overview if there is interest in, and substantial use of, the pricing system.

The convexity adjustment overview consists of two types of convexity adjustments:

- Neglected convexity adjustments;
- Calculated convexity adjustments.

The overview mainly consists of neglected convexity adjustments, because both for Murex and Quark there are a lot of relatively simple pricing functions in use, which do not take convexity adjustments into account. There are some more complex pricing functions in Quark that do take a convexity adjustment into account (e.g., *PMV_Quark_AvgCapFloor*, *PMV_Quark_RateAdj2*).

The outline of this document is as follows. First, in Section 3, the tooling is discussed which PMV uses to create the convexity adjustment overview. Next, in Section 4, it is explained which pricing systems are considered to be in scope for the convexity adjustment overview. Furthermore, the completeness of the convexity adjustment overview is discussed. Then, in Section 5, the concept

of convexity adjustments is discussed in detail. Some specific examples will be given. Section 6 is dedicated to provide updates on the validation deliverables regarding convexity adjustments. Next, in Section 7, it is presented what kind of information is incorporated in the convexity adjustment overview. After that, in Section 8, the focus will be on topics that relate to convexity adjustments that PMV is not able to address at this stage. Finally, Section 9, summarizes the main results and conclusions that have been obtained for this review.

3 Tooling

In this section, the tooling that is used by PMV to create the convexity adjustment overview is discussed.

In order to create and maintain the convexity adjustment overview, the `PMV Model Inventory Application` is used. Details regarding the `PMV Model Inventory Application` can be found in [14].

To give some insight into how the `PMV Model Inventory Application` is used with respect to the convexity adjustment overview, several screenshots of the Graphical User Interface (GUI) are shown below. The *Home* screen is depicted in Figure 3.1.

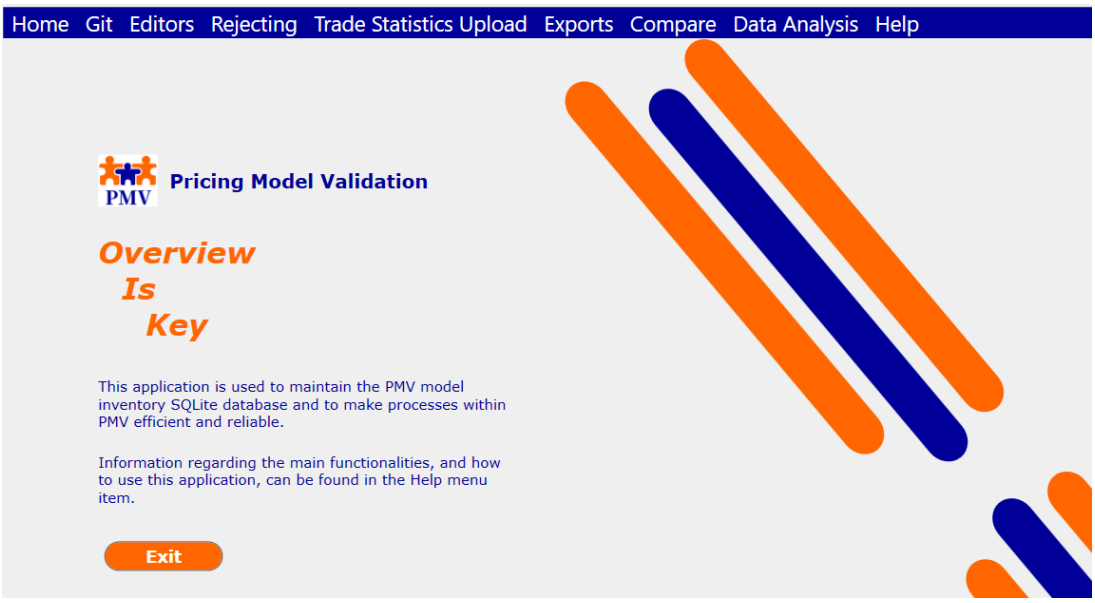

Figure 3.1: The `PMV Model Inventory Application` *Home* screen.

The specific information of all convexity adjustments can be filled in via a form which is depicted in Figure 3.2.

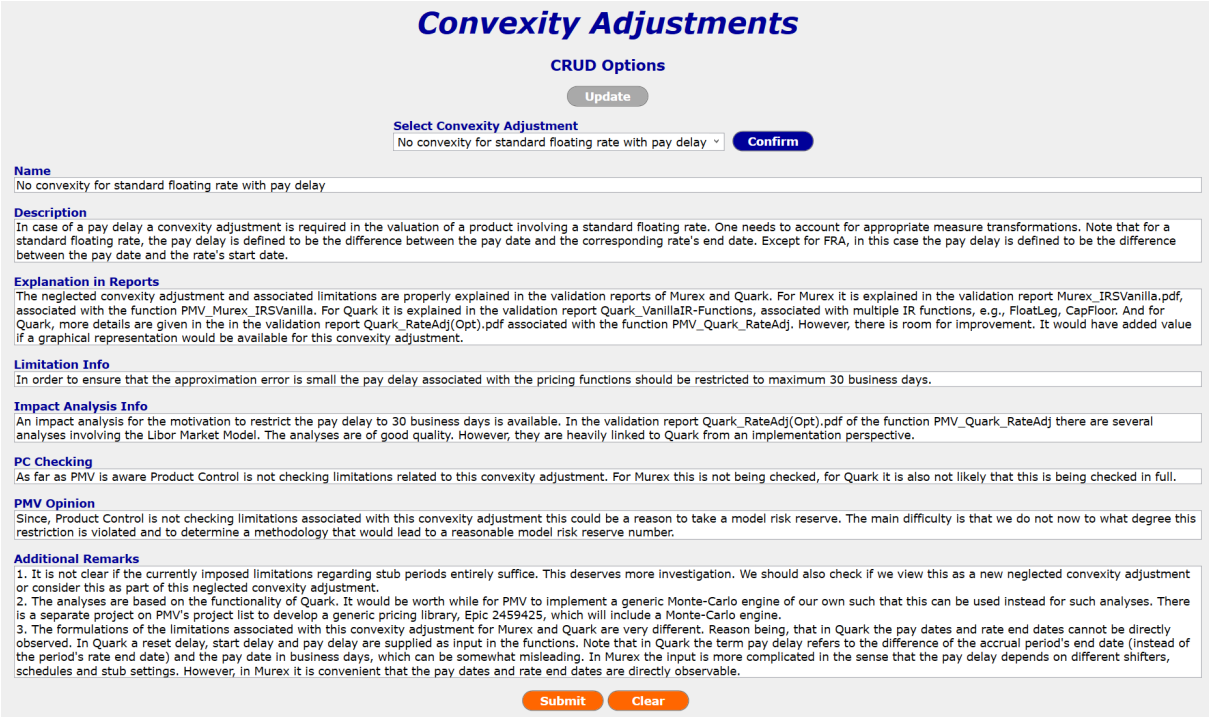

Figure 3.2: The `PMV Model Inventory Application` *Convexity Adjustments* properties screen.

The convexity adjustment can be mapped to pricing functions using the GUI which is shown in Figure 3.3.

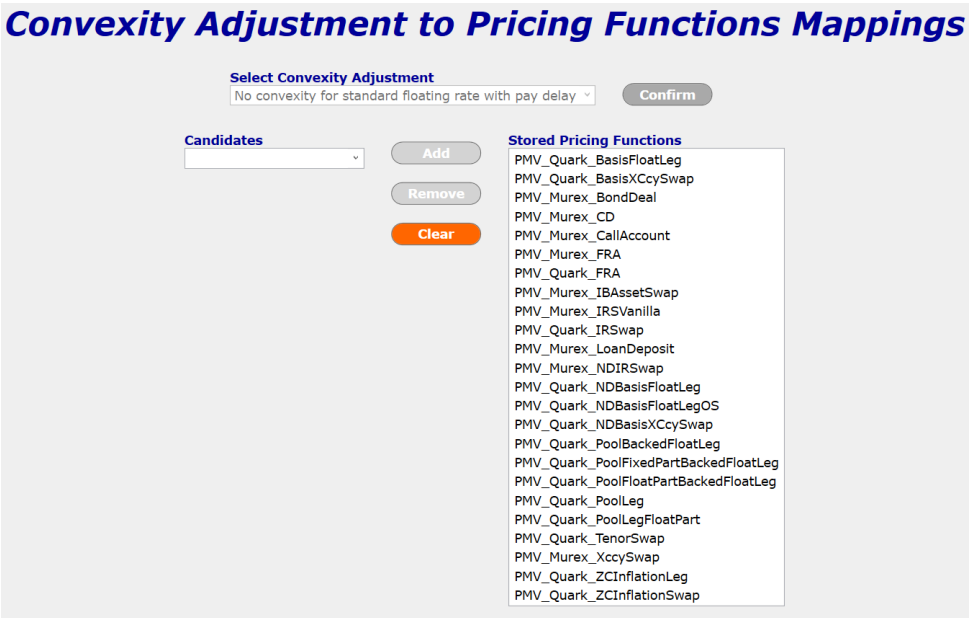

Figure 3.3: The `PMV Model Inventory Application` convexity adjustments to pricing functions mappings screen.

All information regarding convexity adjustments and the mapping to pricing functions is stored in database tables. To get the full overview an Excel export file can be created, *ConvexityAdjust-*

ments.xlsx [12]. The Excel file can be created via the GUI as is shown in Figure 3.4.

Figure 3.4: The `PMV Model Inventory Application` Official Exports screen.

4 Scope & Completeness

In this section it is explained which pricing systems are in scope for the convexity adjustments overview, and if the overview is complete.

During the convexity adjustment periodic review, PMV needs to assess which pricing systems are in scope for the convexity adjustments overview, only pricing systems for which this has added value should be in scope. Furthermore, it needs to be checked if the convexity adjustment overview is up-to-date for the pricing systems that are in scope.

4.1 Murex

Murex is the pricing system, used for P&L, that will eventually replace most pricing systems and is the system for which PMV is getting many new validation requests. Therefore, the (neglected) convexity adjustments of Murex are included in the overview.

For the pricing system Murex, the convexity adjustments are available in the overview since the 2021 periodic review.

The overview has been checked and it turned out that it was up-to-date.

4.2 Quark

It is expected that the pricing system Quark, which is used for P&L, will continue to see a lot of usage in the future. Therefore, the (neglected) convexity adjustments of Quark are included in the overview.

For the pricing system Quark, the convexity adjustments are available in the overview since the 2022 periodic review.

The overview has been checked and it turned out that it was up-to-date.

4.3 Other Systems

Besides Murex and Quark, there are other pricing systems in use at Rabobank. It is even possible that PMV would get a validation request for pricing functions of a completely new pricing system. Though, this is unlikely because Rabobank is in the process of deprecating most pricing systems and instead use Murex. Still, it could happen in practice that PMV is requested to validate pricing functions for a new system which neglect or calculate convexity adjustments.

Currently, the following pricing systems besides Murex and Quark are in use: QuIC and RAM+. The pricing system Sophis and Luna were decommissioned in Q2 2024. The pricing system Sophis was used for P&L purposes, mainly for commodity products. All Sophis trades have been migrated to Murex. The pricing system Luna was used for regulatory calculations in Brazil.

It was already concluded in the periodic review of 2022 that interest to make improvements for these pricing systems within Rabobank, compared to Murex and Quark, is low:

- *QuIC*: The pricing system QuIC is used for the calculation of credit exposures. In QuIC several simplifying design choices and approximations are made, e.g., volatility skew is not included, simplified approach for handling yield curves, use of proxy pricers. As such, the calculation errors resulting from neglecting convexity adjustments are minor compared to the errors that are introduced by these other approximations. QuIC might be replaced by Murex in the future.
- *RAM+*: The single pricing function validated by PMV in the pricing system RAM+ is used to calculate the Initial Margin. There are no convexity adjustments associated with this pricing function.

4.4 Issues

Summary of the issues and actions:

Issue 1 It needs to be checked if the convexity adjustment overview is up-to-date for the pricing systems that are in scope.

Action 1 Firstly, Sophis and Luna have been taken out of the list of considered pricing systems that are in scope for PMV since these pricing system are no longer in use. Secondly, the overview has been checked and it turned out that it was not fully up-to-date. For a Murex neglected convexity adjustment related to average floating rate several fields required to be updated as a result of work done in this review. The comprehensive testing done related to this convexity adjustment is more clearly described in the *IRSVanilla* [13] validation report, and this is now also reflected in the fields of the overview. Furthermore, for Quark some minor typos were fixed. The overview has been updated such that it is up-to-date. The issue has been resolved, but needs to be revisited during next year's periodic review.

5 Convexity Adjustments Explained

In this section, the concept of convexity adjustments will be discussed in detail. Also, some specific examples will be given.

5.1 General

A stochastic variable (or underlying) has a corresponding underlying measure, often referred to as the natural measure. If the expectation of a stochastic variable is calculated under a different measure than the natural measure this gives rise to a so-called convexity adjustment. Formally a convexity adjustment can be defined as below.

Definition (Convexity Adjustment): A convexity adjustment is the difference between the expectation of the (function of the) stochastic variable under a measure, different from the natural measure, and the expectation of the (function of the) stochastic variable under the natural measure.

For the pricing of financial derivatives, typically expectations of (functions of) stochastic rates are calculated.

One well known convexity adjustment in the literature is the CMS convexity adjustment [9]. A fixed-for-floating constant maturity swap (CMS) is a swap, where, in contrast to a plain vanilla swap, the floating leg payments are based on swap/CMS rates rather than on XIBOR rates. For the pricing of CMS-based products,[1] it is necessary to calculate the expectation of the future swap rates under the forward measure that is associated with the pay date. However, the natural martingale measure of the swap rate is the annuity measure. As such, the CMS convexity adjustment is the difference between the expectation of the (function of the) swap rate under the forward measure that is associated with the pay date and the expectation of the (function of the) swap rate under the annuity measure.

Even for much simpler products than CMS-based products, e.g., plain vanilla swaps, cross currency swaps and interest rate futures, convexity adjustments do apply. In the literature, convexity adjustments for these simpler derivatives are not commonly considered. Reason being, that they are often looked at from a simplified view concerning the dates involved when pricing the derivative.

[1] At this stage Murex is not yet used for CMS-based products, but Quark is.

For efficiency purposes, it is desirable to have simple numerical calculations, preferably analytical, in order to calculate the Net Present Value (NPV) and sensitivities for derivatives. To do so, often additional approximations are necessary. These additional approximations can only be taken if the required accuracy is still achieved. The type of additional approximations of interest is the approximation of neglecting a convexity adjustment. A convexity adjustment is a non-linear term in the valuation. By neglecting a convexity adjustment, a risk-factor is removed from the valuation, since the impact of volatility and/or correlation is being ignored. However, the impact of doing so can be negligible under specific conditions. When giving sign-off for pricing functions that can be used to calculate the NPV and sensitivities for derivatives these specific conditions are key. Typically, PMV imposes limitations such that these specific conditions are met. Meaning that within the limitations, neglecting a convexity adjustment is justifiable.

In the next sections a detailed look is taken into some (neglected) convexity adjustments and limitations that are in place for the pricing systems Murex and Quark from a mathematical perspective. These should be seen as examples, PMV's working assumption is that all (neglected) convexity adjustments should be explained in validation reports.

In order to explain convexity adjustments in detail, the focus will be on pricing a plain vanilla interest rate swap (also referred to a fixed-for-floating interest rate swap) and calculating the CMS convexity adjustment.

5.2 Example - Convexity Adjustment due to Delays

In order to explain the concept of a convexity adjustment due to delays, the pricing of an interest rate swap will be considered.

Remark: The notation that will be introduced might initially appear to be more elaborate than required. However, during the 2021 periodic review it was identified that the currently used notation in the validation reports of Murex is too simplistic to properly handle a convexity adjustment that is being neglected due to non-standard rate conventions. This is explained in detail in Section 5.3.

5.2.1 Preliminaries & Notation

We assume that we work in a multi-curve framework. There is one discount curve, this curve is the classic risk-neutral curve (usually we use the Overnight Index Swap (OIS) curve). The price of the zero-coupon bond (ZCB) at time t maturing at time T_i from the discount curve is denoted by $P^D(t, T_i)$.

Additionally, there are forecasting curves which are used to estimate forward rates. A forecasting curve corresponds to an index with a specific tenor. The price of the ZCB at time t maturing at time T_i from the forecasting curve is denoted by $P^F(t, T_i)$.

The forward rate for a period i, denoted by $L_i(t; \mathcal{T}_i^s, \mathcal{T}_i^e)$, is defined as follows:

$$L_i(t) \equiv L(t, \mathcal{T}_i^s, \mathcal{T}_i^e) = \frac{1}{\alpha_i^F} \left(\frac{P^F(t, \mathcal{T}_i^s)}{P^F(t, \mathcal{T}_i^e)} - 1 \right), \tag{5.1}$$

where t denotes the observation date of the forward rate, $\mathcal{T}_i^s$ is start date of the forward rate for

period i, $\mathcal{T}_i^e$ is end date of the forward rate for period i and $\alpha_i^F = \alpha^F(T_i^s, T_i^e)$ denotes the accrual fraction associated with the forecasting curve.

The dates T_i^s and T_i^e are only used to calculate day-counts and can differ from the dates $\mathcal{T}_i^s, \mathcal{T}_i^e$.

The value of an interest rate swap today, denoted by $V_*(t_0)$, is given by the sum of expected cash flows of the two underlying legs. We will refer to the first leg by $*1$ and to the second leg by $*2$.

The following notation will be used for a cash flow:

$$C_i^*(T_i^p), \text{ with } i = 1, \ldots, I,$$

where T_i^p denotes the pay date for the i-th cash flow. We will denote an expected cash flow as follows:

$$\widetilde{C}_i^*(T_i^p), \text{ with } i = 1, \ldots, I.$$

In order to avoid notational confusion in the formulas for the discounted payoffs of the respective legs, the following notation will be used for the indices of the expected cash flows for the first leg $*1$ and second leg $*2$:

$$\widetilde{C}_m^{*1}(T_m^p), \text{ with } m = 1, \ldots, M,$$
$$\widetilde{C}_n^{*2}(S_n^p), \text{ with } n = 1, \ldots, N.$$

So, we make use of the indices m and n. We used letter S instead of T for the second leg flows, as the dates of the two legs can differ.

Furthermore, the symbols $*$, $*1$ and $*2$ will be replaced in the formulas with the cash flow type where applicable, e.g., flt, for floating. The $*$ symbol will also be replaced in $V_*(t_0)$ for the swap type, e.g., swap.

In all our formula's we will look from the perspective of the first leg. From the first leg receiver perspective we have that the value of an interest rate swap today is given by:

$$V_*(t_0) = \sum_{m=1}^{M} \widetilde{C}_m^{*1}(T_m^p) - \sum_{n=1}^{N} \widetilde{C}_n^{*2}(S_n^p),$$

where $\widetilde{C}_m^{*1}(T_m^p)$ denotes the m-th expected cash flow of leg $*1$ and $\widetilde{C}_n^{*2}(S_n^p)$ denotes the n-th expected cash flow of leg $*2$. On the other hand, from the first leg payer perspective we have:

$$V_*(t_0) = \sum_{n=1}^{N} \widetilde{C}_n^{*2}(S_n^p) - \sum_{m=1}^{M} \widetilde{C}_m^{*1}(T_m^p).$$

When the fixed leg is paid the swap is usually called a payer swap, when the fixed leg is received the swap is called a receiver swap. We introduce indicator function ω to indicate if we are dealing with a receiver or payer swap and write:

$$V_*(t_0) = \omega \sum_{m=1}^{M} \widetilde{C}_m^{*1}(T_m^p) - \omega \sum_{n=1}^{N} \widetilde{C}_n^{*2}(S_n^p), \tag{5.2}$$

where ω is given by:

$$\omega = \begin{cases} 1, & \text{if receiver,} \\ -1, & \text{if payer.} \end{cases} \tag{5.3}$$

5.2.2 Fixed & Floating Expected Cash Flows

We will now focus our attention on expected cash flows. An expected cash flow $\widetilde{C}^*(T_i^p)$ under the risk-neural measure corresponding to the numéraire $\mathcal{B}(T_i^p)$ is given by:

$$\widetilde{C}_i^*(T_i^p) = \mathbb{E}_{t_0}^{\mathbb{Q}}\left[\frac{\mathcal{B}(t_0)}{\mathcal{B}(T_i^p)} C_i^*(T_i^p)\right]. \tag{5.4}$$

We can rewrite (5.4) by changing to the T_i^p-forward measure corresponding to the numéraire $P^D(t, T_i^p)$ using Girsanov's theorem:

$$\begin{aligned}\widetilde{C}_i^*(T_i^p) &= \mathbb{E}_{t_0}^{T_i^p}\left[\frac{\mathcal{B}(t_0)}{\mathcal{B}(T_i^p)} \frac{\mathcal{B}(T_i^p)P^D(t_0, T_i^p)}{\mathcal{B}(t_0)P^D(T_i^p, T_i^p)} C_i^*(T_i^p)\right] \\ &= P^D(t_0, T_i^p) \cdot \mathbb{E}_{t_0}^{T_i^p}\left[C_i^*(T_i^p)\right] \\ &= DF^D(T_i^p) \cdot \mathbb{E}_{t_0}^{T_i^p}\left[C_i^*(T_i^p)\right],\end{aligned} \tag{5.5}$$

where $P^D(\cdot,\cdot)$ and $DF^D(\cdot)$ denote the ZCB price and discount factor associated with the discount curve.

Note that in practice, discount factors are used in analytical pricing formulas. Therefore, we will write the analytical pricing formulas in this document, in terms of discount factors rather than ZCB prices.

From (5.5) it follows that we can rewrite (5.2) in the following way:

$$V_*(t_0) = \omega \sum_{n=1}^{N} DF^D(T_n^p) \cdot \mathbb{E}_{t_0}^{T_n^p}[C_n^{*1}(T_n^p)] - \omega \sum_{m=1}^{M} DF^D(T_m^p) \cdot \mathbb{E}_{t_0}^{T_m^p}[C_m^{*2}(T_m^p)]. \tag{5.6}$$

We will throughout this document refer to formula (5.6) as the general formula for the value of a swap today from the perspective of the first leg.

Equations (5.2) and (5.6) take into consideration that the frequency of the two underlying legs can differ, resulting in the fact that one leg can have more pay dates than the other.

Fixed Cash Flows

The cash flows for the fixed leg of an interest rate (IR) swap are referred to as fixed cash flows and involve a fixed rate. The fixed cash flow that takes place at pay date T_i^p, denoted by $C_i^{\text{fxd}}(T_i^p)$, is defined as follows:

$$C_i^{\text{fxd}}(T_i^p) = Q \cdot \alpha_i^{\text{fxd}} \cdot K_i, \tag{5.7}$$

where Q denotes the notional, $\alpha_i^{\text{fxd}} = \alpha^{\text{fxd}}(T_i^s, T_i^e)$ denotes the accrual fraction of the fixed leg and K_i denotes the fixed rate.

We will now determine a pricing formula for the expected fixed cash flow $\widetilde{C}_i^{\text{fxd}}(T_i^p)$. From (5.5) it follows that the expected fixed cash flow $\widetilde{C}_i^{\text{fxd}}(T_i^p)$ under the T_i^p-forward measure is given by:

$$\widetilde{C}_i^{\text{fxd}}(T_i^p) = DF^D(T_i^p) \cdot \mathbb{E}_{t_0}^{T_i^p}\left[C_i^{\text{fxd}}(T_i^p)\right]. \tag{5.8}$$

Note that we can calculate the expectation on the RHS of (5.8) exactly, since the i-th fixed cash flow is known at pay date T_i^p. As such, we have:

$$E_{t_0}^{T_i^p}[C_i^{\text{fxd}}(T_i^p)] = Q \cdot \alpha_i^{\text{fxd}} \cdot K_i. \tag{5.9}$$

Hence, the expected fixed cash flows are given by:

$$\widetilde{C}_i^{\text{fxd}}(T_i^p) = DF^D(T_i^p) \cdot C_i^{\text{fxd}}(T_i^p). \tag{5.10}$$

Floating Cash Flows

The cash flows for the floating leg of an IR swap are referred to as floating cash flows and involve a floating rate. The floating cash flow that takes place at pay date T_i^p, denoted by $C_i^{\text{flt}}(T_i^p)$ is defined as follows:

$$C_i^{\text{flt}}(T_i^p) = Q \cdot \alpha_i^{\text{flt}} \cdot (L_i(T_i^r) + \mathcal{M}_i), \tag{5.11}$$

where Q denotes the notional, $\alpha_i^{\text{flt}} = \alpha^{\text{flt}}(T_i^s, T_i^e)$ denotes the accrual fraction of the floating leg, $\mathcal{M}_i$ denotes the margin, T_i^r denotes the reset/fixing date of the forward rate and $L_i(T_i^r) = L_i(T_i^r; \mathcal{T}_i^s, \mathcal{T}_i^e)$ denotes the forward rate between forward start date $\mathcal{T}_i^s$ and forward end date $\mathcal{T}_i^e$ observed at reset date T_i^r taken from the forecasting curve.

Note that now much more dates are involved, compared to the fixed cash flow case. Depending how the dates are set compared to one another can give rise to certain kind of *delays*.

Definition (Delay): A delay is defined to be a closed interval between two dates, the size of the interval will be determined by the number of days between the two dates. Here, days can be business days or calendar days, although typically business days are used. We distinguish 4 different delays[2] that are defined as follows:

- The *reset delay*, denoted by $\mathcal{R}$, is the closed interval between the fixing date (also known as the reset date) T_i^r and the start date of the accrual period T_i^s.
- The *start delay*, denoted by θ, is the closed interval between the reset date T_i^r and the start date of the forward rate $\mathcal{T}_i^s$.
- The *pay delay*, denoted by $\mathcal{P}$ is the closed interval between the forward rate end date $\mathcal{T}_i^e$ and the pay date T_i^p.
- The *Quark pay delay*, denoted by $\bar{\mathcal{P}}$ is the closed interval between the end date of the accrual period T_i^e and the pay date T_i^p. This is as the name suggest a delay that is specifically used within Quark. As such, it is only relevant for Quark pricing functions. In Quark, the pay delay is not used. Instead the Quark pay delay, together with other delays are used to obtain the desired date schedule.

Furthermore, we define η to be the difference of the reset delay and the start delay:

$$\eta = \mathcal{R} - \theta. \tag{5.12}$$

We will consider two examples, a simple example and a realistic example, in order to get better insight in the role of the underlying dates for the floating leg of an interest rate swap.

Example 1: Simple - no delays.

In this simple example, the following assumptions are made with respect to the underlying dates of the floating leg:

[2] There are more delays, e.g., FX reset delay.

- The start dates of the accrual periods, the start date of the forward rates and the reset dates coincide:

$$T_i^s = T_i^r = \mathcal{T}_i^s, \text{ for } i = 1, \ldots, I.$$

As a result there is no reset delay and no start delay:

$$\mathcal{R} = 0b,$$
$$\theta = 0b.$$

- The end dates of the accrual periods, the end dates of the forward rates and the payment dates coincide:

$$T_i^e = T_i^p = \mathcal{T}_i^e, \text{ for } i = 1, \ldots, I.$$

As a result there are no pay delays:

$$\bar{\mathcal{P}} = 0b,$$
$$\mathcal{P} = 0b.$$

Also note that the start date of the accrual period coincides with the end date of the previous accrual period (exception being the first accrual period, since their is no previous end date):

$$T_{i+1}^s = T_i^e, \text{ for } i = 1, \ldots, I-1.$$

The dates and cash flows for the floating leg are depicted for $I = 2$ in Figure 5.1.

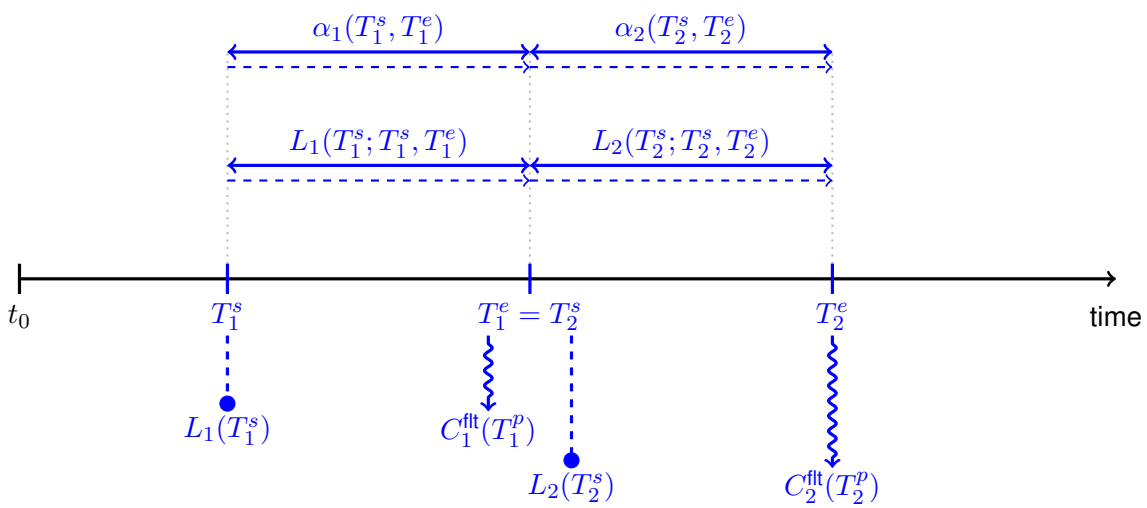

Figure 5.1: Simple example: dates and cash flows for a floating leg.

Example 2: Realistic - dates and coupon payments for the floating leg.
In this realistic example the following assumptions are made with respect to the underlying dates of the floating leg:

- The start dates of the accrual periods, the start date of the forward rates and the fixing rates do not coincide. More precisely the following holds:

$$T_i^r < \mathcal{T}_i^s < T_i^s, \text{ for } i = 1, \ldots, I.$$

As a result there is a reset delay and a start delay, for example:

$$\mathcal{R} = 4b,$$
$$\theta = 3b,$$
$$\eta = 1b.$$

- The end dates of the accrual periods, the end dates of the forward rates and the payment dates do not coincide. More precisely the following holds:

$$\mathcal{T}_i^e < T_i^e < T_i^p, \text{ for } i = 1, \ldots, I.$$

As a result there are pay delays:

$$\bar{\mathcal{P}} = 7b,$$
$$\mathcal{P} = 8b.$$

Also, note that the start date of the accrual period coincides with the end date of the previous accrual period and the start date of the forward rate coincides with the end date of the forward rate of the previous forward rate period (exceptions being the first periods, since their are no previous end dates):

$$T_{i+1}^s = T_i^e, \text{ for } i = 1, \ldots, I-1,$$
$$\mathcal{T}_{i+1}^s = \mathcal{T}_i^e, \text{ for } i = 1, \ldots, I-1.$$

The dates and cash flows for the floating leg are depicted for $I = 2$ in Figure 5.2.

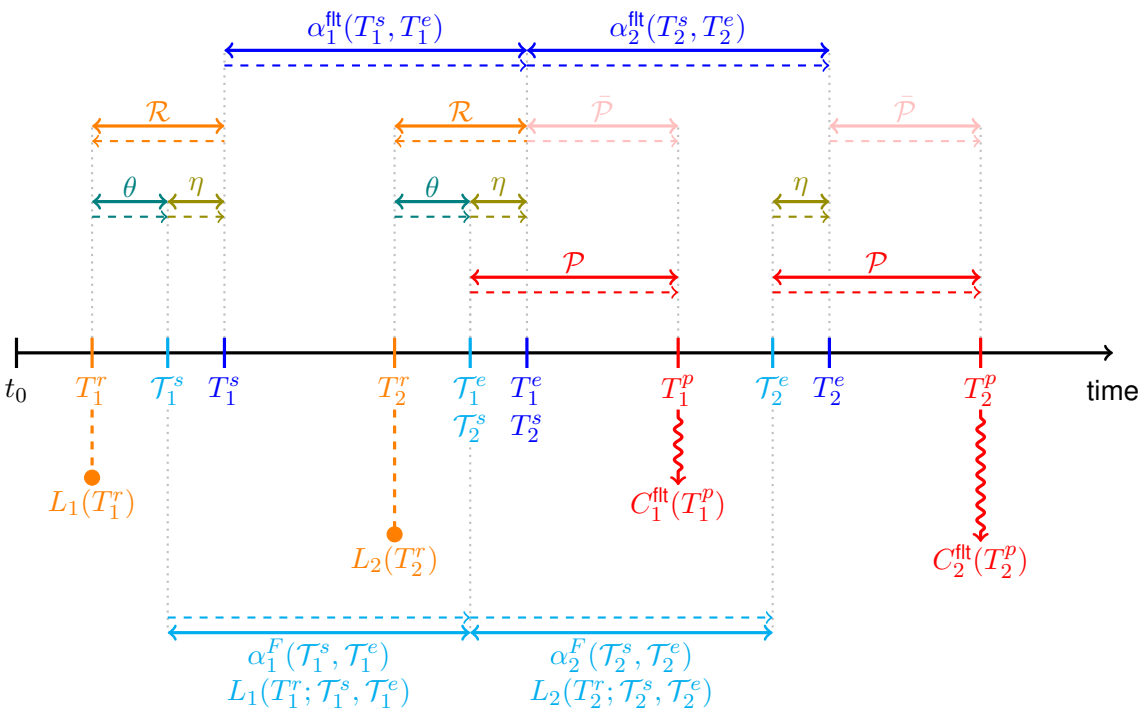

Figure 5.2: Realistic example: dates and cash flows for a floating leg.

Remark: The explanation of the delays and their graphical representations is currently not available in a validation report. The topic to improve this has been included in the convexity adjustment overview.

We will now determine a pricing formula for the expected floating cash flow $\widetilde{C}_i^{\text{flt}}(T_i^p)$ and we will refer back to Figure 5.2 as it will turn out that the pay delay $\mathcal{P}$ plays an important role.

From (5.5) it follows that the expected floating cash flow $\widetilde{C}_i^{\text{flt}}(T_i^p)$ under the T_i^p-forward measure is given by:

$$\widetilde{C}_i^{\text{flt}}(T_i^p) = DF^D(T_i^p) \cdot \mathbb{E}_{t_0}^{T_i^p}\left[C_i^{\text{flt}}(T_i^p)\right]. \tag{5.13}$$

From (5.11) it follows that:

$$E_{t_0}^{T_i^p}\left[C_i^{\text{flt}}\left(T_i^p\right)\right] = \mathbb{E}_{t_0}^{T_i^p}\left[Q \cdot \alpha_i^{\text{flt}} \cdot \left(L_i(T_i^r) + \mathcal{M}_i\right)\right]. \tag{5.14}$$

The expectation on the RHS of (5.14) cannot be calculated exactly, however the expectation can be approximated by making use of an additional assumption.

We can rewrite (5.14) as follows:

$$\begin{aligned}
\mathbb{E}_{t_0}^{T_i^p}\left[C_i\left(T_i^p\right)\right] &= Q \cdot \alpha_i^{\text{flt}} \cdot \left(\mathbb{E}_{t_0}^{T_i^p}\left[L_i(T_i^r; \mathcal{T}_i^s, \mathcal{T}_i^e)\right] + \mathcal{M}_i\right) && (5.15)\\
&\approx Q \cdot \alpha_i^{\text{flt}} \cdot \left(\mathbb{E}_{t_0}^{\mathcal{T}_i^e}\left[L_i(T_i^r; \mathcal{T}_i^s, \mathcal{T}_i^e)\right] + \mathcal{M}_i\right) && (5.16)\\
&= Q \cdot \alpha_i^{\text{flt}} \cdot \left(L_i(t_0; \mathcal{T}_i^s, \mathcal{T}_i^e) + \mathcal{M}_i\right) && (5.17)\\
&= Q \cdot \alpha_i^{\text{flt}} \cdot \left(\frac{1}{\alpha_i^F}\left(\frac{P^F(t_0, \mathcal{T}_i^s)}{P^F(t_0, \mathcal{T}_i^e)} - 1\right) + \mathcal{M}_i\right) && (5.18)\\
&= Q \cdot \frac{\alpha_i^{\text{flt}}}{\alpha_i^F} \cdot \left(\left(\frac{DF^F(\mathcal{T}_i^s)}{DF^F(\mathcal{T}_i^e)} - 1\right) + \mathcal{M}_i\right). && (5.19)
\end{aligned}$$

Note that 1 additional approximation is used:

- In (5.16), the convexity adjustment associated with changing from the T_i^p-forward measure to the $\mathcal{T}_i^e$-forward measure is neglected.

Furthermore, the following was applied to derive formula (5.19):

- In (5.15) the terms that are known, are taken out of the expectation.
- In (5.17) we use that the fact that $T_i^r < \mathcal{T}_i^e$ and that the forward rate $L_i(\mathcal{T}_i^e)$ is a martingale under the $\mathcal{T}_i^e$-forward measure.
- In (5.18) we substitute the definition of the forward rate (5.1).
- In (5.19) we express the ZCBs in terms of discount factors.

Hence, the expected floating cash flows are given by:

$$\widetilde{C}_i^{\text{flt}}(T_i^p) = DF^D(T_i^p) \cdot Q \cdot \frac{\alpha_i^{\text{flt}}}{\alpha_i^F} \cdot \left(\left(\frac{DF^F(\mathcal{T}_i^s)}{DF^F(\mathcal{T}_i^e)} - 1\right) + \mathcal{M}_i\right). \tag{5.20}$$

The key to obtain analytical formula (5.20) is neglecting the convexity adjustment associated with changing from the T_i^p-forward measure to the $\mathcal{T}_i^e$-forward measure. This change in measure is directly related to the pay delay. As previously mentioned the pay delay $\mathcal{P}$ denotes the closed

interval between the forward rate end date $\mathcal{T}_i^e$ and the pay date T_i^p which can also be seen in Figure 5.2. Neglecting the convexity adjustment, is of course not mathematically sound. However, if the pay date T_i^p is not too far from the forward rate end date $\mathcal{T}_i^e$ the effect of neglecting the convexity adjustment will be sufficiently small.

In [11] it is shown by means of an impact analysis that neglecting the convexity adjustment is acceptable up to a pay delay of $30b$, $\mathcal{P} = 30b$. So this means that the cash flow or coupon payment date T_i^p should not be more than 30 business days away from the forward rate end date $\mathcal{T}_i^e$ for each period i. As such, the following limitation is defined, *a period's pay date cannot be more than 30 business days away from the corresponding period's rate end date*.

Remark: The analysis in [11] is done using a Quark pricing function. The way dates are set in Quark and Murex, functionality wise is very different. The limitations that are set for Quark functions are in general based on direct inputs of the Quark pricing function. The reset delay (and cash delay,[3] although technically this is obtained from the forecasting curve), Quark pay delay and start delay (this relates to backward looking rates) are used to set the relevant dates. In Quark 3 limitations (one for the reset delay, one for the Quark pay delay and one for the start delay) are imposed in order to achieve that the pay date T_i^p is not more than 30 business days away from the forward rate end date $\mathcal{T}_i^e$ for each period i. The main underlying reason is that in Quark the forward rate end date $\mathcal{T}_i^e$ is not directly observable, in Murex this date is observable. As such, in Murex 1 limitation suffices.

Plain Vanilla Swap Pricing Formula

Now that the fixed and expected cash flows are determined, the pricing formula can be given for a plain vanilla swap. The value of a plain vanilla swap, today will be denoted by $\mathrm{V}_{\text{swap}}(t_0)$. Remember that the general formula for the value of a swap today from the perspective of the first leg is given by (5.6). In case of a plain vanilla fixed-for-floating swap, the first leg $*1$ has fixed cash flows and the second leg $*2$ has floating cash flows:

$$C_m^{*1}(T_m^p) = C_m^{\text{fxd}}(T_m^p),$$
$$C_n^{*2}(T_n^p) = C_n^{\text{flt}}(T_n^p).$$

As such, (5.6) can be written as:

$$\mathrm{V}_{\text{swap}}(t_0) = \omega \sum_{m=1}^{M} DF^D(T_m^p) \cdot E_{t_0}^{T_m^p}[C_m^{\text{fxd}}(T_m^p)] - \omega \sum_{n=1}^{N} DF^D(T_n^p) \cdot \mathbb{E}_{t_0}^{T_n^p}[C_n^{\text{flt}}(T_n^p)],$$

where ω is given by (5.3), $E_{t_0}^{T_m^p}[C_m^{\text{fxd}}(T_m^p)]$ is given by (5.9) and $\mathbb{E}_{t_0}^{T_n^p}[C_n^{\text{flt}}(T_n^p)]$ is given by (5.19).

5.3 Example - Convexity Adjustment due to Non-Standard Rate Conventions

For Murex, PMV discovered that a neglected convexity adjustment is currently not discussed in any validation report, this concerns the item *No convexity for floating rates with non-standard rate conventions*.

The definition of the forward rate in (5.1) is actually incomplete as this only concerns the most standard linear case.

[3]The cash delay is specified via the curve. The start date of the forward date is obtained by rolling the reset date forward by the cash delay.

The forward rate depends on the underlying rate convention.[4] How interest accrues over time within Murex is determined via the `RateConvention` object, see [7]. A rate convention consists out of 2 main components:

1. The basis according to which a measure of time is computed;
2. The computing mode which indicates which accrual formula is used.

There are 5 different rate conventions that can be used:

1. *Linear - Standard basis*;
2. *Linear - Discount basis*;
3. *Linear - FRA basis*;
4. *Yield*;
5. *Exponential*.

See [10] and [7] for more details.

The *Linear - Standard basis* rate convention is considered to be a standard rate convention, the rest are considered to be non-standard rate conventions.

Taking the different rate conventions into account, the forward rate for a period i, denoted by $L_i(t;\mathcal{T}_i^s,\mathcal{T}_i^e)$, is defined as follows:

$$L_i(t,\mathcal{T}_i^s,\mathcal{T}_i^e) = \begin{cases} \frac{1}{\alpha_i^F}\left(\frac{P^F(t,\mathcal{T}_i^s)}{P^F(t,\mathcal{T}_i^e)} - 1\right), & \text{if } \texttt{RateConvention} = \text{Linear - Standard basis}, \\ \frac{1}{\alpha_i^F}\left(1 - \frac{P^F(t,\mathcal{T}_i^e)}{P^F(t,\mathcal{T}_i^s)}\right), & \text{if } \texttt{RateConvention} = \text{Linear - Discount basis}, \\ \frac{1}{\alpha_i^F}\left(\frac{P^F(t,\mathcal{T}_i^e)}{P^F(t,\mathcal{T}_i^s)-P^F(t,\mathcal{T}_i^e)} - 1\right), & \text{if } \texttt{RateConvention} = \text{Linear - FRA basis}, \\ \left(\frac{P^F(t,\mathcal{T}_i^s)}{P^F(t,\mathcal{T}_i^e)}\right)^{\frac{1}{\alpha_i^F}} - 1, & \text{if } \texttt{RateConvention} = \text{Yield}, \\ \frac{1}{\alpha_i^F}\log\left(\frac{P^F(t,\mathcal{T}_i^s)}{P^F(t,\mathcal{T}_i^e)}\right), & \text{if } \texttt{RateConvention} = \text{Exponential}. \end{cases} \tag{5.21}$$

The floating cash flow that takes place at pay date T_i^p, denoted by $C_i^{\text{flt}}(T_i^p)$, also depends on the underlying rate convention and is defined as follows:

$$C_i^{\text{flt}}(T_i^p) = \begin{cases} Q\cdot\alpha_i^{\text{flt}}\cdot\left(L_i(T_i^r)+\mathcal{M}_i\right), & \text{if } \texttt{RateConvention} = \text{Linear - Standard basis}, \\ Q\cdot\frac{\alpha_i^{\text{flt}}\cdot\left(L_i(T_i^r)+\mathcal{M}_i\right)}{1-\alpha_i^{\text{flt}}\cdot\left(L_i(T_i^r)+\mathcal{M}_i\right)}, & \text{if } \texttt{RateConvention} = \text{Linear - Discount basis}, \\ Q\cdot\frac{1}{1+\alpha_i^{\text{flt}}\cdot\left(L_i(T_i^r)+\mathcal{M}_i\right)}, & \text{if } \texttt{RateConvention} = \text{Linear - FRA basis}, \\ Q\cdot\left((1+L_i(T_i^r)+\mathcal{M}_i)^{\alpha_i^{\text{flt}}} - 1\right), & \text{if } \texttt{RateConvention} = \text{Yield}, \\ Q\cdot\left(e^{\alpha_i^{\text{flt}}\cdot\left(L_i(T_i^r)+\mathcal{M}_i\right)} - 1\right), & \text{if } \texttt{RateConvention} = \text{Exponential}. \end{cases} \tag{5.22}$$

[4] A rate convention is more Murex terminology, although the expectation is that generally speaking the same will hold true for other pricing systems, e.g., Quark.

We will now focus our attention to expected floating cash flows.

Before, we derived an analytical pricing formula for the expected floating cash flows for the standard rate convention *Linear - Standard basis*. We will now do the same for the non-standard *Yield* rate convention to show that there are additional assumptions necessary compared to the standard rate convention to obtain an analytical pricing formula.

Remark: We will only consider the *Yield* rate convention, but the same holds true for the majority of the other non-standard rate conventions.

The expected floating cash flows in case of the *Yield* rate convention is given by:

$$\begin{aligned}
\mathbb{E}_{t_0}^{T_i^p}\left[C_i\left(T_i^p\right)\right] &= \mathbb{E}_{t_0}^{T_i^p}\left[Q\cdot\left(\left(1+L_i(T_i^r)+\mathcal{M}_i\right)^{\alpha_i^{\text{flt}}}-1\right)\right] && (5.23)\\
&= Q\cdot\left(\mathbb{E}_{t_0}^{T_i^p}\left[\left(1+L_i(T_i^r)+\mathcal{M}_i\right)^{\alpha_i^{\text{flt}}}\right]-1\right) && (5.24)\\
&= Q\cdot\left(\mathbb{E}_{t_0}^{T_i^p}\left[\left(1+\left(\frac{P^F(T_i^r,\mathcal{T}_i^s)}{P^F(T_i^r,\mathcal{T}_i^e)}\right)^{\frac{1}{\alpha_i^F}}-1+\mathcal{M}_i\right)^{\alpha_i^{\text{flt}}}\right]-1\right) && (5.25)\\
&\approx Q\cdot\left(\mathbb{E}_{t_0}^{T_i^p}\left[\left(\frac{P^F(T_i^r,\mathcal{T}_i^s)}{P^F(T_i^r,\mathcal{T}_i^e)}\right)^{\frac{\alpha_i}{\alpha_i^F}}\right]-1\right) && (5.26)\\
&\approx Q\cdot\left(\mathbb{E}_{t_0}^{T_i^p}\left[\frac{P^F(T_i^r,\mathcal{T}_i^s)}{P^F(T_i^r,\mathcal{T}_i^e)}\right]-1\right) && (5.27)\\
&\approx Q\cdot\left(\mathbb{E}_{t_0}^{\mathcal{T}_i^e}\left[\frac{P^F(T_i^r,\mathcal{T}_i^s)}{P^F(T_i^r,\mathcal{T}_i^e)}\right]-1\right) && (5.28)\\
&= Q\cdot\left(\frac{P^F(t_0,\mathcal{T}_i^s)}{P^F(t_0,\mathcal{T}_i^e)}-1\right) && (5.29)\\
&= Q\cdot\left(\frac{DF^F(\mathcal{T}_i^s)}{DF^F(\mathcal{T}_i^e)}-1\right). && (5.30)
\end{aligned}$$

Note that 3 additional approximations are used:

- In (5.26) the margin $\mathcal{M}_i$ is assumed to be equal to zero, $\mathcal{M}_i = 0$.
- In (5.27) the accrual fraction of the floating leg is assumed to be equal to the accrual fraction associated with the underlying forecasting curve, $\alpha_i^{\text{flt}} = \alpha_i^F$.
- In (5.28) the convexity adjustment associated with changing from the T_i^p-forward measure to the $\mathcal{T}_i^e$-forward measure is neglected.

Furthermore, the following was applied to derive formula (5.30):

- In (5.24) the terms that are known, are taken out of the expectation.
- In (5.25) we substitute the definition of the forward rate for the Yield rate convention, (5.21).
- In (5.29) we use that the fact that $T_i^r < \mathcal{T}_i^e$ and that the forward rate $L_i(\mathcal{T}_i^e)$ is a martingale under the $\mathcal{T}_i^e$-forward measure.
- In (5.30) we express the ZCBs in terms of discount factors.

Remark: In this example we have only considered standard floating rates. There are similar issues for compounded and averaged floating rates.

Hence, the expected floating cash flows in case of the *Yield* rate convention are given by:

$$\widetilde{C}_i^{\text{flt}}(T_i^p) = DF^D(T_i^p) \cdot Q \cdot \left(\frac{DF^F(\mathcal{T}_i^s)}{DF^F(\mathcal{T}_i^e)} - 1 \right). \tag{5.31}$$

Note that two additional assumptions were necessary to obtain analytical formula (5.31), besides neglecting the convexity adjustment associated with changing from the T_i^p-forward measure to the $\mathcal{T}_i^e$-forward measure. Additionally, assumptions were needed regarding the margin and the accrual fraction. In practice though, a margin of zero and equality of the accrual fraction of the floating leg and the accrual fraction associated with the underlying forecasting curve will not always hold. More investigation is still needed to come up with appropriate limitations.

This issue is handled further via a separate project on PMV's project list, Epic 2190248 and is still ongoing. It is quite a large project as it involves a complete rewrite of the *IRSVanilla* [13] validation report and updating several settings tables. The only way to handle this issue is via a complete rewrite. Reason being, that new notations, derivations and analyses are required before new limitations can be imposed. It is also expected that other validation reports will eventually require a change with respect to the used notation. Currently, this is on hold as another validation project is currently being worked on that is linked to the same validation report that requires updating, Epic 8426332. The work will be resumed once this validation is finished.

For now part of this information is still included in this specific report of the convexity adjustment overview. However, the idea is to eventually remove this content from this report when the rewrite of the *IRSVanilla* validation report is finished.

5.4 Example - CMS Convexity Adjustment

As previously mentioned, in Section 5.1, one well known convexity adjustment in the literature is the CMS convexity adjustment [9]. There are a variety of CMS-based products actively traded in the market. CMS-based products involve swap rates.

We denote by $S(t; T_a, T_b)$, the swap rate observed at time t of an underlying swap that starts at time T_a and matures at time T_b. The swap rate is defined as follows:

$$S(t; T_a, T_b) = S_{T_a,T_b}(t) = \frac{P(t, T_a) - P(t, T_b)}{A_{a,b}(t)}, \tag{5.32}$$

where the annuity factor or annuity is given by:

$$A(t) = A_{a,b}(t) = \sum_{i=1}^{N} P(t, T_i)\alpha_i. \tag{5.33}$$

The simplest CMS-based products are CMS swaps, caps and floors.

Remark: Currently Rabobank is trading CMS swaps, caps and floors. Trades can be booked via the use of the Quark functions, PMV_Quark_FloatLeg and PMV_Quark_CapFloor.

For an N-period CMS swap, $T_a = T_{i-1}$ and $T_b = T_{i-1+N}$. In case of a fixed-for-floating N-period CMS swap, the first leg $*1$ has fixed cash flows and the second leg $*2$ has floating cash flows in which the floating rate is indexed by a reference swap rate:

$$C_m^{*1}(T_m^p) = C_m^{\text{fxd}}(T_m^p),$$
$$C_n^{*2}(T_n^p) = C_n^{\text{CMS}}(T_n^p).$$

As such, (5.6) can be written as:

$$\mathrm{V}_{\text{CMS}}(t_0) = \omega \sum_{m=1}^{M} DF^D(T_m^p) \cdot E_{t_0}^{T_m^p}[C_m^{\text{fxd}}(T_m^p)] - \omega \sum_{n=1}^{N} DF^D(T_n^p) \cdot \mathbb{E}_{t_0}^{T_n^p}[C_n^{\text{CMS}}(T_n^p)],$$

where ω is given by (5.3), $E_{t_0}^{T_m^p}[C_m^{\text{fxd}}(T_m^p)]$ is given by (5.9) and $\mathbb{E}_{t_0}^{T_n^p}[C_n^{\text{CMS}}(T_n^p)]$ is given by:

$$\begin{aligned}\mathbb{E}_{t_0}^{T_i^p}[C_i^{\text{CMS}}(T_i^p)] &= \mathbb{E}_{t_0}^{T_i^p}\left[Q \cdot \alpha_i^{\text{CMS}} \cdot S(T_{i-1}; T_{i-1}, T_{i-1+N})\right] \\ &= Q \cdot \alpha_i^{\text{CMS}} \cdot \mathbb{E}_{t_0}^{T_i^p}\left[S(T_{i-1}; T_{i-1}, T_{i-1+N})\right]. \end{aligned} \tag{5.34}$$

The swap rate under the T_p-forward measure is commonly referred to as a CMS rate. The expectation in (5.34) is not straightforward to calculate. Reason being, that the swap rate is not a martingale under the T_p-forward measure. The natural measure of a swap rate is the annuity measure that has the annuity as the numéraire.

Since, we are only interested in the CMS convexity adjustment we will focus our attention on a single CMS-linked cash flow for the swap rate,

$$S(T_0) = S_{0,N}(T_0) = S(T_0; T_0, T_N).$$

The value of the CMS-linked cash flow is defined by:

$$V_{\text{gCMS}}(t_0) = \mathbb{E}_{t_0}^{T_p}\left[g(S(T_0))\right], \tag{5.35}$$

where

$$g(s) = \begin{cases} s, & \text{in case of a swaplet,} \\ (s-K)^+, & \text{in case of a caplet,} \\ (K-s)^+, & \text{in case of a floorlet.} \end{cases} \tag{5.36}$$

Here K denotes the strike. Note that a CMS swap is a collection of CMS-linked cash flows with $g = s$. Similarly, CMS caps/floors are a collection of CMS-linked cash flows with $g = (s-K)^+$ or $g = (K-s)^+$.

Let us be clear on the notations, consider the case of a swaplet, then:

- t_0 denotes today's date.
- T_0 denotes the start date of the underlying swap (e.g., 1 year from now). This will also be the start of the accrual period of the swaplet.
- T_p denotes the pay date of the swaplet (e.g., 6 months from T_0). This will also be the end of the accrual period of the swaplet.
- T_N denotes the maturity date of the underlying swap (e.g., 10 years from T_0).

- T_r denotes the fixing/reset date of the swap rate. The reset date is usually two days before the start date of the underlying swap. For simplicity we will ignore the reset date, $T_r = T_0$.

We know that for a continuous random variable Y and any function $h(\cdot)$ the expectation of $h(Y)$ is defined as:

$$\mathbb{E}[h(Y)] = \int_{-\infty}^{\infty} h(y)\psi(y)dy, \tag{5.37}$$

where $\psi(\cdot)$ is the Probability Density Function (PDF) of Y. So, for the expectation in (5.35) we can write:

$$\mathbb{E}_{t_0}^{T_p}\left[g(S(T_0))\right] = \int_{-\infty}^{\infty} g(s)\psi^{T_p}(s)\mathrm{d}s, \tag{5.38}$$

where $\psi_p^T(\cdot)$ is the PDF of a swap rate in the T_p-forward measure. However, the PDF $\psi_p^T(\cdot)$ is not directly available. The PDF $\psi^A(\cdot)$ of a swap rate in the annuity measure on the other hand can be obtained from the market prices of swaptions [8, 2] via

$$\psi^A(x) = \begin{cases} \frac{\partial^2 c(t_0,S(t_0),T_0,x)}{\partial x^2}, & \text{if } x \geq S(t_0), \\ \frac{\partial^2 p(t_0,S(t_0),T_0,x)}{\partial x^2}, & \text{if } x < S(t_0), \end{cases} \tag{5.39}$$

where

$$c(t_0, S(t_0), T_0, x) = \mathbb{E}_{t_0}^A\left[(S(T_0) - x)^+\right], \tag{5.40}$$

$$p(t_0, S(t_0), T_0, x) = \mathbb{E}_{t_0}^A\left[(x - S(T_0))^+\right]. \tag{5.41}$$

Under the annuity measure, a payer swaption can be interpreted as a call option on a swap rate, while a receiver swaption can be interpreted as a put option on a swap rate.

From the market we can imply the dynamics of $S(T_0)$ in the annuity measure, we can rewrite (5.35) by changing to the annuity measure corresponding to the numéraire $A(t)$ using Girsanov's theorem:

$$V_{\text{gCMS}}(t_0) = \mathbb{E}_{t_0}^A\left[g(S(T_0))\frac{P(T_0,T_p)A(t_0)}{P(t_0,T_p)A(T_0)}\right] \tag{5.42}$$

$$= \frac{A(t_0)}{P(t_0,T_p)}\mathbb{E}_{t_0}^A\left[\frac{P(T_0,T_p)}{A(T_0)}g(S(T_0))\right]. \tag{5.43}$$

The CMS convexity adjustment is given by:

$$\Lambda_{\text{gCMS}}(t_0) = \mathbb{E}_{t_0}^{T_p}\left[g(S(T_0))\right] - \mathbb{E}_{t_0}^A\left[g(S(T_0))\right]. \tag{5.44}$$

The difficulty in calculating the expectation in (5.43) stems from the term $P(T_0,T_p)/A(T_0)$, since it depends on the joint distribution of a whole set of interest rates. In order to compute this expectation, a term-structure model can be used.

Though, it is also possible to make use of a good analytical approximation via the use of a Terminal Swap Rate (TSR) model [9, 5]. In a TSR model, the term $P(T_0,T_p)/A(T_0)$ is approximated with a so-called *annuity mapping function*, denoted by $\gamma(S(T_0))$. Different TSR models can be used for this purpose. The most popular TSR model is the swap-yield TSR model, the annuity function for this TSR model is given by:

$$\gamma(s) = \frac{s}{1 - \prod_{i=1}^{N}\frac{1}{1+\alpha_i s}}. \tag{5.45}$$

When using an annuity mapping function, expression (5.43) can be rewritten as follows:

$$V_{\text{gCMS}}(t_0) = \frac{A(t_0)}{P(t_0,T_p)}\mathbb{E}_{t_0}^A\left[\gamma(S(T_0))g(S(T_0))\right], \tag{5.46}$$

where $\gamma(\cdot)$ is an annuity mapping function.

The expectation in (5.46) can be calculated using the *replication method*. The replication method is used to replicate the CMS payout by means of European swaptions of various strikes. This method is very popular in practice [6, 3] because it takes the volatility smile effects into account. Therefore, it is sometimes referred to as the street-standard model for the CMS convexity adjustment. For more information about the replication method itself, please consult [4].

To apply the replication method, equation (5.46) first needs to be rewritten:

$$V_{\text{gCMS}}(t_0) = \frac{A(t_0)}{P(t_0, T_p)} \mathbb{E}_{t_0}^{A} \left[f(S(T_0)) \right], \tag{5.47}$$

where $f(S(T_0)) = \gamma(S(T_0))g(S(T_0))$. We can write the expectation as an integral over the PDF:

$$\mathbb{E}^{A} \left[f(S(T_0)) \right] = \int_{-\infty}^{\infty} f(x)\psi^{A}(x)\mathrm{d}x, \tag{5.48}$$

where $\psi^{A}(\cdot)$ is given in (5.39).

We are now ready to apply the replication method. Let $f(\cdot)$ be defined on the interval $[a, b]$.[5] The calculation of $\mathbb{E}^{A} \left[f(S(T_0)) \right]$ in (5.48) is subdivided into 3 different cases depending on the value of the swap rate, $S(t_0)$, compared to the boundary conditions, a and b:

1. If $S(t_0) < a$:

$$\begin{aligned}\mathbb{E}_{t_0}^{A} \left[f(S(T_0)) \right] = {} & f(b)\frac{\partial c(t_0, S(t_0), T_0, b)}{\partial x} - f(a)\frac{\partial c(t_0, S(t_0), T_0, a)}{\partial x} \\ & - f'(b)c(t_0, S(t_0), T_0, b) + f'(a)c(t_0, S(t_0), T_0, a) \\ & + \int_a^b f''(x)c(t_0, S(t_0), T_0, x)\mathrm{d}x.\end{aligned}$$

2. If $S(t_0) > b$:

$$\begin{aligned}\mathbb{E}_{t_0}^{A} \left[f(S(T_0)) \right] = {} & f(b)\frac{\partial p(t_0, S(t_0), T_0, b)}{\partial x} - f(a)\frac{\partial p(t_0, S(t_0), T_0, a)}{\partial x} \\ & - f'(b)p(t_0, S(t_0), T_0, b) + f'(a)p(t_0, S(t_0), T_0, a) \\ & + \int_a^b f''(x)p(t_0, S(t_0), T_0, x)\mathrm{d}x.\end{aligned}$$

3. If $a \leq S(t_0) \leq b$:

$$\begin{aligned}\mathbb{E}_{t_0}^{A} \left[f(S(T_0)) \right] = {} & f(S(t_0)) - f(a)\frac{\partial p(t_0, S(t_0), T_0, a)}{\partial x} + f(b)\frac{\partial c(t_0, S(t_0), T_0, b)}{\partial x} \\ & + f'(a)p(t_0, S(t_0), T_0, a) - f'(b)c(0, S(0), T_0, b) \\ & + \int_a^{S(t_0)} f''(x)p(t_0, S(t_0), T_0, x)\mathrm{d}x + \int_{S(t_0)}^{b} f''(x)c(t_0, S(t_0), T_0, x)\mathrm{d}x,\end{aligned}$$

where $p(t_0, S(t_0), T_0, x)$ and $c(t_0, S(t_0), T_0, x)$ are defined by (5.40) and (5.41).

The minimum strike, K_{min}, or maximum strike, K_{max}, are chosen based on numerical considerations. The values of the boundaries a, b differ depending on the type of CMS option.

[5] This is for numerical reasons.

- In case of a CMS caplet: $a = K$, where K is the given strike, and $b = K_{\max}$.
- In case of a CMS floorlet: $b = K$, where K is the given strike, and $a = K_{\min}$.
- In case of a CMS swaplet: $a = K$, where $K_{\min} = K = 0$ and $b = K_{\max}$.

From the above it can be concluded that calculating a CMS convexity adjustment is quite involved, and that the calculated CMS convexity adjustment is model dependent.

5.5 Issues

In principle no issues in this periodic review are mentioned regarding the convexity adjustments themselves, since that is precisely what the convexity adjustment overview is for. However, for the *No convexity for floating rates with non-standard rate conventions* item we make an exception. Reason being, that this was one of the main drivers to create the convexity adjustment overview and the fact that currently this issue is not reflected in any validation report.

Summary of the issues and actions:

Issue 2 One neglected convexity adjustment, *No convexity for floating rates with non-standard rate conventions*, deserves special attention as it has not been handled during any validation or periodic review thus far. This item is part of the convexity adjustment overview but it is not explained within any Murex validation report, nor were any limitations imposed.

Action 2 The neglected convexity adjustment, *No convexity for floating rates with non-standard rate conventions*, was already added to the convexity adjustments overview during the periodic review of 2021. For now these explanations and derivations are included in this specific report. However, the idea is to move these explanations and derivations to the validation report of the pricing function *IRSVanilla* [13]. The rewrite of this validation report is a separate project on PMV's project list, Epic 2190248 and is still ongoing. Currently, another validation project is being worked on wherein part of the rewrite will be done, Epic 8426332. However, the expectation is that after this validation more work will still be required as the rewrite will include adding comprehensive testing for some neglected convexity adjustments that could also result in imposing additional limitations. This issue remains open.

6 Updates to Validation Deliverables

The convexity adjustment overview shows that for some convexity adjustments, the validation deliverables are lacking (e.g., explanation in a validation report is incomplete, impact analysis is missing).

In the periodic review of 2023 PMV looked into detail at several comprehensive tests in validation reports. It was observed that the main issue is the fact that the underlying files that are used to create the comprehensive tests are not maintained in an effective way. Most are not under version control and more importantly it is difficult to recreate comprehensive tests for new input data. Therefore, PMV created a new repo better maintain all materials related to comprehensive testing. The idea of this repo is that it will be as self-contained as possible, such that it easy to recreate comprehensive

tests. PMV is working on moving tests to this new repo, which is still a substantial amount of work. The progress is tracked via a separate project on PMV's project list, Epic 8192740.

6.1 Issues

Summary of the issues and actions:

Issue 3 The convexity adjustment overview shows that for some convexity adjustments, the validation deliverables are lacking (e.g., explanation in a validation report is incomplete, impact analysis is missing).

Action 3 In the periodic review of 2023 a new repo was created to better maintain all materials related to comprehensive testing. The idea of this repo is that it will be as self-contained as possible, such that it is easy to recreate comprehensive tests. In this year PMV has worked on the design of the repo and put a structure in place. PMV also moved two comprehensive tests to this repo, one for Quark and one for Murex. Moving all comprehensive tests is still a substantial amount of work. The expectation is that with the structure in place this work can now also be done as part of ongoing related validations. The progress for the comprehensive testing repo is tracked via a separate project on PMV's project list, Epic 8192740. Furthermore in this review, PMV moved the performed analysis of the 2022 periodic review on Convexity Adjustment for Average Indices (Murex) to the *IRSVanilla* [13] validation report, and improved the testing section in this validation report. This issue remains open.

7 Convexity Adjustments Overview Information

In this section some information will be given regarding the convexity adjustment overview.

7.1 Excel File

In Section 3 it was explained that the convexity adjustments overview can be exported as a *ConvexityAdjustments.xlsx* Excel file [12]. This Excel file has two worksheets:

- **Info**: the *Info* worksheet gives an explanation of the content that can be found in the *Main* worksheet.
- **Main**: the *Main* worksheet is the actual overview of all the convexity adjustments, this sheet has several columns.

The following columns can be found in the convexity adjustment overview.

- **Name**: name of the (neglected) convexity adjustment.
- **Description**: description of the (neglected) convexity adjustment.

- **Pricing System**: list of pricing systems for which this convexity adjustment is relevant.
- **Explanation in Reports**: information regarding if the explanation in the validation reports is sufficient.
- **Limitation Info**: information regarding the limitations associated with this convexity adjustment.
- **Impact Analysis Info**: information regarding if impact analyses are available and what the status is of these analyses.
- **PC Checking**: information regarding if Product Control is checking limitations associated with this convexity adjustment.
- **PMV Opinion**: opinion of PMV regarding the status of this convexity adjustment.
- **Additional Remarks**: additional remarks that are relevant.
- **Pricing Function**: list of pricing functions for which this convexity adjustment is relevant.

7.2 Number of Convexity Adjustments

In Table 7.1 a high level overview is given of the number of convexity adjustments per pricing system as of November 2024.

Pricing System	# Neglected CAs	# Calculated CAs	# Total CAs
Murex Only	7	1	8
Quark Only	1	3	4
Murex & Quark	10	0	10
All	18	4	22

Table 7.1: Overview of the number of convexity adjustments (CAs) per pricing system as of November 2024. Convexity adjustments can apply to a single or multiple pricing systems.

In Table 7.1 it can be seen that there are quite some (neglected) convexity adjustments associated with pricing functions, in total 22 convexity adjustments. Only 4 calculated convexity adjustments, and 18 neglected convexity adjustments. Of the 18 neglected convexity adjustments, 7 apply only to Murex, 1 applies only to Quark, and the remaining 10 apply to both Murex and Quark. These counts are subject to change in the future, since more pricing functions can be requested for validation and pricing functions can also be retired. Furthermore, change in insight could lead to redefine the currently formulated convexity adjustments.

7.3 Examples

To give some more insight, the information of 4 convexity adjustments that have been included in the overview for different reasons are shown:

- *No convexity for floating rates with non-standard rate conventions*;
- *Calculated convexity adjusted CMS rate for CMS-based products*;

- *No convexity for CtD discounting*;
- *No convexity for standard floating rate with pay delay*.

Remark: The information for the fields of each specific item are presented to reflect what is shown in the Excel overview.

7.3.1 No convexity for floating rates with non-standard rate conventions

This neglected convexity adjustment is currently not reflected in any validation report, this is one of the most important issues to fix with respect to convexity adjustments for signed-off pricing functions. This neglected convexity adjustment applies only to Murex. The following information is stored in the overview:

- **Name**: No convexity for floating rates with non-standard rate conventions.
- **Description**: The forward rate depends on the underlying rate convention, the rate convention indicates how interest accrues over time. Calculating expectations for floating rates (e.g., standard, compounded, averaged) which involve a non-standard rate convention requires incorporating a convexity adjustment in the pricing engine.
- **Pricing System**: Murex.
- **Explanation in Reports**: This item is currently not explained in any validation report, this should be added. The most logical place would be the validation report Murex_IRSVanilla.pdf of the function PMV_Murex_IRSVanilla.
- **Limitation Info**: Currently there are no limitations. The expectation is that limitations have to be imposed regarding the margin and the accrual fraction of the floating leg and the accrual fraction associated with the forecasting curve.
- **Impact Analysis Info**: Not Applicable.
- **PC Checking**: Not Applicable.
- **PMV Opinion**: This observed issue was actually the main driver to start with the convexity adjustments review. In order to handle this properly the idea is to rewrite the validation report Murex_IRSVanilla.pdf of the function PMV_Murex_IRSVanilla, the current notation and structure in the report are not suitable to address this issue. As this is quite a large task (also other issues in this document need to be addressed), a separate project has been created on PMV's project list, epic number: 2190248. The idea is that the validation report will become one of the main documents that other validation reports can refer to.
- **Additional Remarks**: 1. In Murex there are 5 different rate conventions: (1) Linear - Standard basis, (2) Linear - Discount basis, (3) Linear - FRA basis, (4) Yield, (5) Exponential. Linear - Standard basis is considered to be a standard rate convention, the rest are considered as non-standard. 2. It is at this stage not fully clear how many functions this concerns. The ones that PMV believes are in scope have been mapped but this still needs more investigation. 3. Eventually it should be checked to what degree this is properly covered in Quark, the expectation is that this topic is also not covered in full.
- **Pricing Function**: PMV_Murex_CD; PMV_Murex_FRA; PMV_Murex_IRSVanilla; PMV_Murex_LoanDeposit; PMV_Murex_NDIRSwap; PMV_Murex_NDLoanDeposit; PMV_Murex_XccySwap.

7.3.2 Calculated convexity adjusted CMS rate for CMS-based products

This is a calculated convexity adjustment that only applies to Quark. There are at this stage no reasons to make any improvements with respect to this convexity adjustment. The following information is stored in the overview:

- **Name**: Calculated convexity adjusted CMS rate for CMS-based products.
- **Description**: When pricing CMS-based products involving swap rates the price needs to be calculated under the payment measure. However, the natural measure for a swap rate is the annuity measure. Under the annuity measure the swap rate is a martingale. A terminal swap rate (TSR) model can be used to analytically approximate a swap rate under the payment measure. The resulting rate is commonly referred to as a (convexity adjusted) CMS rate. The swap-yield TSR model is used is used to calculate convexity adjusted CMS rates in production.
- **Pricing System**: Quark.
- **Explanation in Reports**: The swap-yield TSR model which is used to calculate convexity adjusted CMS rates is properly explained in the validation report Quark_RateAdj(Opt)2.pdf associated with the PMV_Quark_RateAdj2 function. The pricing of CMS-based derivatives is explained in the validation reports Quark_CMSSpreadCapFloor(Digital).pdf and Quark_SpdOptSmile.pdf.
- **Limitation Info**: No limitations are imposed, since this is a calculated convexity adjustment.
- **Impact Analysis Info**: There are multiple impact analyses available, in the validation report Quark_RateAdj(Opt)2.pdf associated with the PMV_Quark_RateAdj2 function, to judge the quality of the used swap-yield TSR model in production:

  1. There are multiple comparisons between the used TSR model used in production vs a Libor Market Model.
  2. There are multiple comparisons between the used TSR model used in production vs other TSR models.

  These analyses are all of good quality.
- **PC Checking**: Since no limitations are formulated, no checking is required by Production Control regarding the limitations. Furthermore, it deserves to be mentioned that Product Control performs Independent Verification (IPV) Testing for CMS-bases products.
- **PMV Opinion**: No action is required regarding this calculated convexity adjustment, the use of a swap-yield TSR model to calculate convexity adjusted CMS rates can be considered to be a market standard approach.
- **Additional Remarks**:

  1. Currently Rabobank is trading CMS and CMS spread derivatives.
  2. For the pricing of CMS spread derivatives, specific functions are available (e.g., PMV_Quark_CMSSpreadCapFloor). For the pricing of CMS derivatives, the pricing functions are used PMV_Quark_FloatLeg and PMV_Quark_CapFloor are used. Note that these 2 functions are broadly used, their implementation is more generic. They are also used for pricing vanilla legs.
  3. The pricing of CMS spread derivatives is more complex than the pricing of CMS swaps. A TSR model is used to calculate convexity adjusted CMS rates. However, other modelling assumptions, are also used in order to price CMS spread derivatives.

4. SpreadOptSmile not adjusted yet to work with RFR rates. FEng is working on this and will send PMV a validation request when ready.

- **Pricing Function**: PMV_Quark_CMSSpreadCapFloor; PMV_Quark_CMSSpreadDigital; PMV_Quark_CapFloor; PMV_Quark_FloatLeg; PMV_Quark_RateAdj2; PMV_Quark_RateAdjOpt2; PMV_Quark_SpdOptSmile.

7.3.3 No convexity for CtD discounting

This is a neglected convexity adjustment that applies to a lot of pricing functions of both Murex and Quark. Neglecting this convexity adjustment is completely justified as it is considered to be market standard. There are at this stage no reasons to make any improvements with respect to this convexity adjustment. The following information is stored in the overview:

- **Name**: No convexity for CtD discounting.
- **Description**: In the Cheapest-to-Deliver (CtD) framework it is assumed that FX forward rates are independent from the collateral currency such that it is possible to extend the regular and cross-currency adjusted discount factors to discount factors in any other currency, this is a market standard assumption. In reality convexity adjustments need to be applied to derive a formula for the cross-currency adjusted collateralized zero coupon bond. However, calibrating these potential convexity adjustments is currently not possible as relevant market data is not available.
- **Pricing System**: Murex; Quark.
- **Explanation in Reports**: The neglected convexity adjustment is properly explained in the validation reports of Murex and Quark. For Murex it is explained in the validation report Murex_RateCurveFramework.pdf, associated with the function PMV_Murex_RateCuves. For Quark it is explained in the validation report Quark_FxFramework.pdf, associated with the function PMV_Quark_BuildFX (note that there are multiple FxObject functions). There are no limitations imposed as incorporating this convexity adjustment in the pricing engine is not market standard as there is no relevant market data available for calibrating these convexity adjustments.
- **Limitation Info**: No limitations are imposed.
- **Impact Analysis Info**: There are no impact analyses available. This is also not necessary at this stage, due to the lack of market data the only feasible analysis would be an artificial experiment which would have little added value.
- **PC Checking**: Not Applicable.
- **PMV Opinion**: No action is required regarding neglecting this convexity adjustment. This will change if relevant market data would become available, which would be picked up during periodic review or be handled via a new validation.
- **Additional Remarks**: No additional remarks.
- **Pricing Function**[6]: PMV_Quark_AvgFloatLeg; PMV_Quark_BasisFloatLeg; PMV_Quark_CMM-AutoCap; PMV_Quark_CSMM-Digital2D; PMV_Quark_FRA; PMV_Murex_FRA; PMV_Murex_FXSwap; PMV_Quark_FXSwap; PMV_Murex_IRSVanilla; PMV_Murex_NDIRSwap.

[6] Only a subset of functions is shown as there are many functions for which this neglected convexity adjustment applies. For the full list of functions, the *ConvexityAdjustments.xlsx* file should be consulted.

7.3.4 No convexity for standard floating rate with pay delay

This is a neglected convexity adjustment that applies to a lot of pricing functions of both Murex and Quark. The following information is stored in the overview:

- **Name**: No convexity for standard floating rate with pay delay.
- **Description**: In case of a pay delay a convexity adjustment is required in the valuation of a product involving a standard floating rate. One needs to account for appropriate measure transformations. Note that for a standard floating rate the pay delay is defined to be the difference between the pay date and the corresponding rate's end date. Except for FRA, in this case the pay delay is defined to be the difference between the pay date and the rate's start date.
- **Pricing System**: Murex; Quark.
- **Explanation in Reports**: The neglected convexity adjustment and associated limitations are properly explained in the validation reports of Murex and Quark. For Murex it is explained in the validation report Murex_IRSVanilla.pdf, associated with the function PMV_Murex_IRSVanilla. For Quark it is explained in the validation report Quark_VanillaIR-Functions, associated with multiple IR functions, e.g., PMV_Quark_FloatLeg, PMV_Quark_CapFloor. And for Quark, more details are given in the in the validation report PMV_Quark_RateAdj(Opt)2.pdf associated with the function PMV_Quark_RateAdj2. However, there is room for improvement. It would have added value if a graphical representation would be available for this convexity adjustment.
- **Limitation Info**: In order to ensure that the approximation error is small the pay delay associated with the pricing functions should be restricted to maximum 30 business days.
- **Impact Analysis Info**: An impact analysis for the motivation to restrict the pay delay to 30 business days is available. In the validation report Quark_RateAdj(Opt)2.pdf of the function PMV_Quark_RateAdj2 there are several analyses involving the Libor Market Model. The analyses are of good quality. However, they are heavily linked to Quark from an implementation perspective.
- **PC Checking**: As far as PMV is aware, Product Control is not checking limitations related to this convexity adjustment. For Murex this is not being checked, for Quark it is also not likely that this is being checked in full.
- **PMV Opinion**: Since, Product Control is not checking limitations associated with this convexity adjustment this could be a reason to take a model risk reserve. The main difficulty is that we do not now to what degree this restriction is violated and to determine a methodology that would lead to a reasonable model risk reserve number.
- **Additional Remarks**:
  1. It is not clear if the currently imposed limitations regarding stub periods entirely suffice. This deserves more investigation. We should also check if we view this as a new neglected convexity adjustment or consider this as part of this neglected convexity adjustment.
  2. The analyses are based on the functionality of Quark. It would be worth while for PMV to implement a generic Monte-Carlo engine of our own such that this can be used instead for such analyses. There is a separate project on PMV's project list to develop a generic pricing library, Epic 2459425, which will include a Monte-Carlo engine.

3. The formulations of the limitations associated with this convexity adjustment for Murex and Quark are very different. Reason being, that in Quark the pay dates and rate end dates cannot be directly observed. In Quark a reset delay, start delay and pay delay are supplied as input in the functions. Note that in Quark the term pay delay refers to the difference of the accrual period's end date (instead of the period's rate end date) and the pay date in business days, which can be somewhat misleading. In Murex the input is more complicated in the sense that the pay delay depends on different shifters, schedules and stub settings. However, in Murex it is convenient that the pay dates and rate end dates are directly observable.

- **Pricing Function**[6]: PMV_Murex_BondDeal; PMV_Murex_CD; PMV_Murex_FRA; PMV_Murex_IRSVanilla; PMV_Murex_NDIRSwap; PMV_Quark_IRSwap; PMV_Quark_PoolBackedFloatLeg; PMV_Quark_TenorSwap.

8 Other Topics

This section should be seen as a placeholder where items are discuss which PMV wants to address in the future, but where PMV is not yet in the position to address these items.

8.1 Model Risk Reserve

As per the Model Risk Reserve policy [1] PMV is responsible for detecting model weaknesses and proposing a model risk reserve, if material. The approximations related to neglecting convexity adjustments are natural candidates for model risk reserves, since these can be regarded as a model weakness. Currently, no model risk reserves are in place related to these model weaknesses.

A complicating factor is that most limitations related to convexity adjustments that have been imposed by PMV, which are a mitigating factor, are currently not being checked by Product Control (PC). Reason being, that for most convexity adjustments, limitations need to be checked for market data and payoff functions. PC is not checking market data for Murex and Quark. Currently, it is unknown to what degree the limitations associated with convexity adjustments are being violated. PC is working on improving the checking of limitations but this is a large topic and the limitations related to convexity adjustments are the most complicated ones. As such, it is not expected that these limitations will be checked in the near future.

Taking a model risk reserve does not absolve PC from their responsibility to properly check the limitations in place. In fact, in PMV's opinion checking these limitations is very important. The models in production are only suitable for use within the signed-off settings, this is precisely the reason why the limitations are imposed.

The current replication libraries of PMV are heavily linked to the design of the pricing systems and their input. Therefore, they are not flexible enough such that they can be easily used for generic analyses. In practice, one would want to have a generic Monte-Carlo framework that could be used to make such generic analyses.

The development of a generic pricing library which will include a generic Monte-Carlo framework is a separate project on PMV's project list, Epic 2459425. This will mainly be developed to improve the

comprehensive testing that is being done by PMV as part of some validations. The hope is that this library can eventually also help with determining whether a model risk reserve is deemed necessary for a particular model. Note that this is a very large project and is currently on hold. It is not clear if it will be worked on next year as the primary focus is currently on the new comprehensive testing repo, Epic 8192740. As such, it is expected that it will still take considerable time before this can be used in the model risk reserve process.

8.2 Issues

Summary of the issues and actions:

Issue 4 The neglecting of convexity adjustments can be viewed as model weaknesses. As per the model risk reserve policy [1], PMV should consider proposing a model risk reserve in case of a model weakness. Currently, no model risk reserves are in place related to these model weaknesses.

Action 4 Most limitations related to convexity adjustments that have been imposed by PMV, which are a mitigating factor, are currently not being checked by Product Control (PC) as far as PMV is aware. Reason being, that PC is not checking market data for Murex and Quark. To mitigate the effect of these violations, a model risk reserve should be taken. PMV planned to create a generic pricing library which will include a generic Monte-Carlo framework. This is a separate project on PMV's project list, Epic 2459425. The hope is that this library can eventually also help with determining whether a model risk reserve is deemed necessary for a particular model. This project is currently on hold and it is not clear if it will be worked on next year as the primary focus is currently on the new comprehensive testing repo, Epic 8192740. This issue remains open.

9 Conclusion

See Executive Summary.

References

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